33 kV FROM HV SWITCHBOARD
F00-SW-010
2000-F00-SLD-EL-10004

F13-SR-001
415V SAG MILL SWITCHROOM

F13-TR-002
TRANSFORMER ROOM

F13-TR-002
33/0.433kV
2500kVA
Dyn11
z=6%
ANAF
NOTE 3

SWR EARTH BAR
Tx EARTH BAR
4000/ 1A
2.5VA
5P20

F13-SW-002
SAG MILL 415V MOTOR CONTROL CENTRE

MCC EARTH BAR
NEUTRAL BAR
NOTE 1
1/700A
4000/ 5A
5VA
5P20
00F-01
4000/ 5A
30VA
C10.5
TEST
PM
20A
63A
2A
32A
240VAC SPRING
CHARGE SUPPLY
00F-08
ACB
AFD
2A
4000A
50G
24VDC CONTROL POWER SUPPLY
NOTE 7
125A
160A
SD
NOTE 6

NORTHERN STAR RESOURCES LIMITED

Approved	X
Approved as Noted	
Not Approved	
For Information Only	

Signed [redacted] Date 15/09/2025

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

415V 3NE~ 50Hz 4000A 65kA for 1sec

4F-01	4F-11	5R-17	3F-01	3F-05	3F-09 INC	1R-05	3F-13	1F-01	1F-05	1F-09	1F-13			
100A	100A	20A	32A	32A	32A	630A	32A	32A	32A	32A	32A			
2A 32A														
100/1A	90A	70A	70A	70A	70A	630A	70A	70A	70A	70A	70A			
		EOL	EOL	EOL	EOL		EOL	EOL	EOL	EOL	EOL			

FEEDER TO F13-SW-102
2000-F13-SCM-EL-10324

E-SHUT
F13SR001-LCS
NOTE 12

CONTINUED ON SHT 2
2000-F13-SLD-EL-10302

37kW	29KW	0.37kW	7.5kW	7.5kW	7.5kW	7.5kW	7.5kW	7.5kW	7.5kW	7.5kW	7.5kW			
F13-CV-404 RECYCLE CONVEYOR 1 2000-F13-SCM-EL-10311	F13CU407-CP SAG MILL VSD COOLING SYSTEM 2000-F13-SCM-EL-10310	F13-FN-425 SAG MILL MOTOR LUBE SKID COOLER FAN 2000-F13-SCM-EL-10312	F13-FN-427 SAG MILL MOTOR 1 FAN 1 2000-F13-SCM-EL-10313	F13-FN-428 SAG MILL MOTOR 1 FAN 2 2000-F13-SCM-EL-10314	F13-FN-429 SAG MILL MOTOR 1 FAN 3 2000-F13-SCM-EL-10315	F13-FN-430 SAG MILL MOTOR 1 FAN 4 2000-F13-SCM-EL-10316	F13-FN-431 SAG MILL MOTOR 2 FAN 1 2000-F13-SCM-EL-10317	F13-FN-432 SAG MILL MOTOR 2 FAN 2 2000-F13-SCM-EL-10318	F13-FN-433 SAG MILL MOTOR 2 FAN 3 2000-F13-SCM-EL-10319	F13-FN-434 SAG MILL MOTOR 2 FAN 4 2000-F13-SCM-EL-10320				

NOTES

1. EARTH-NEUTRAL LINK SHALL BE INSTALLED IN INCOMER TIER. SIGNAGE SHALL BE PLACED ON MCC AND ON TRANSFORMER READING "M.E.N. LINK LOCATED IN F13-SW-002 INCOMER TIER" CABLE SHALL BE IDENTIFIED C/W LABEL STATING "M.E.N. LINK - DO NOT REMOVE".
2. SWITCHBOARD SHALL BE EXTENSIBLE AND BE PROVIDED WITH THE FACILITY TO EXTEND THE MAIN BUSBAR AND APPEND ADDITIONAL TIERS AT ONE END WITHOUT MAJOR STRUCTURAL DISASSEMBLY.
3. UPSTREAM TRANSFORMER FEEDER SHALL BE TRIPPED ON ACTIVATION OF SWITCHROOM EMERGENCY PUSHBUTTON, DETECTION OF EARTH FAULT OR ARC FAULT DETECTION.
4. SWITCHBOARD VENDOR SHALL SELECT SWITCHGEAR TO PROVIDE TYPE-2 CO-ORDINATION SUITABLE FOR HIGH EFFICIENCY MOTORS.
5. ALL DRIVE I/O VIA COMMS NETWORK TO DCS. FOR COMMUNICATIONS BLOCK DIAGRAM REFER 2000-F13-BLK-IC-10002.
6. SURGE DIVERTERS SHALL PROVIDE FEEDBACK TO DCS VIA VOLT-FREE CONTACTS. SURGE DIVERTER FUSES SHALL BE SIZED BY VENDOR TO SUIT DUTY.
7. MINIATURE CIRCUIT BREAKERS SHALL BE SIZED AND RATED TO SUIT ACB AND INCOMER REQUIREMENTS.
8. THERMAL OVERLOAD UNITS MAY BE SUBSTITUTED WITH ELECTRONIC (MAGNETIC) OVERLOAD UNITS TO SUIT DUTY.
9. TOLs TO BE MOUNTED IN PURPOSE-BUILT ENCLOSURE BENEATH VSD. TOLs MAY BE INCLUDED IN MCC MODULE WITH ENGINEERING APPROVAL.
10. ANY DRIVE RTDs SHALL BE WIRED TO DEDICATED REMOTE I/O CARDS IN THE FIELD. RTD RELAYS MAY BE INCLUDED IN THE MCC MODULE WITH ENGINEERING APPROVAL.
11. CIRCUIT BREAKER TO BE SUBSTITUED FOR FUSED-SWITCH WHERE REQUIRED BY EQUIPMENT MANUFACTURERS GUIDELINES.
12. EMERGENCY SHUTDOWN (E-SHUT) PUSH BUTTON N/O WIRED TO SHUNT TRIP NOMINATED CIRCUIT BREAKERS.

LEGEND

- DIRECT ON-LINE MOTOR STARTER GENERIC - NOTE 4, 5
- DIRECT ON-LINE MOTOR STARTER REVERSING - NOTE 4, 5
- EOL — ELECTRONIC OVERLOAD UNIT NOTE 4, 5
- THERMAL OVERLOAD UNIT NOTE 8
- VARIABLE SPEED DRIVE NOTE 4, 5
- EARTH LEAKAGE RELAY NOTE 4, 5
- FIELD-MOUNTED DECONTACTOR
- MOTOR THERMISTOR (ONLY REQ. FOR MOTORS >37kW - 250kW)
- RTD — MOTOR RTD (ONLY REQ. FOR MOTORS >250kW)
- MOTOR HEATERS (ONLY REQ. FOR MOTORS >90kW)
- 49 63 71 — TRANSFORMER PROTECTIVE DEVICES (REFER ANSI STANDARDS FOR DESCRIPTIONS)
- CIRCUIT BREAKER (NOTE 11)
- xxxA ST TRIP SET POINT / xxxA CB SIZE
- AFD — ARC FLASH DETECTION
- ECR — EARTH CONTINUITY RELAY
- 1 E-SHUT — SWITCHROOM EMERGENCY SHUTDOWN
- 63A 3 PHASE SOCKET OUTLET

DRG No.	REFERENCES
2000-F13-SCM-EL-10303	F13-SW-002 415V MCC ARC FLASH SYSTEM SCHEMATIC DIAGRAM
2000-F13-SCM-EL-10302	F13-SW-002 415V MCC 24VDC DISTRIBUTION SCHEMATIC DIAGRAM
2000-F13-SCM-EL-10301	F13-SW-002 415V MCC INCOMER SCHEMATIC DIAGRAM
2000-F13-DRG-EL-10101	F13-SR-001 415V SWITCHROOM ELECTRICAL EQUIPMENT LAYOUT
2000-F13-BLK-IC-10001	F13-SR-001 415V SWITCHROOM AND REMOTE I/O COMMUNICATIONS BLOCK DIAGRAM
2000-F13-SLD-EL-10301	F13-SW-002 415V MCC SINGLE LINE DIAGRAM SHEET 2
2000-F00-SLD-EL-10002	F00-SW-010 33kV SWITCHBOARD SINGLE LINE DIAGRAM SHEET 1 OF 5
2000-F00-STD-EL-10001	GENERAL NOTES AND STANDARD DRAWINGS INDEX

REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP
1	18AUG25	FAT REDLINES INCORPORATED					
0	30MAY24	APPROVED FOR CONSTRUCTION					

NORTHERN STAR RESOURCES LIMITED

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PROJECT: NORTHERN STAR RESOURCES LTD
KCGM GROWTH PROJECT

TITLE: MILLING & CLASSIFICATION - SAG MILL
F13-SW-002 415V MCC
SINGLE LINE DIAGRAM SHEET 1

SHEET	SCALE	DRG No.	REVISION
A1	NONE	2000-F13-SLD-EL-10301	1

O:\CAD PROJECTS\23S NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\ELF13 - MILLING & CLASSIFICATION\2000-F13-SLD-EL-10301DWG 27AUG25 7:50AM HOTSEAT04